

Project Report: AssetPulse - Crypto Portfolio Studio

1. Executive Summary

AssetPulse is a real-time cryptocurrency portfolio management application designed to provide users with an institutional-grade tracking dashboard. By leveraging the **CoinMarketCap API**, the application offers live market data, profit/loss (P/L) analysis, and portfolio allocation visualizations. The project focuses on data privacy by utilizing local storage and ensures security through a custom Python-based proxy server.

2. Technical Architecture

The project follows a decoupled architecture to ensure performance and security:

- **Frontend:** Built with a "Vanilla First" approach using **HTML5**, **CSS3 (Custom Properties & Grid/Flexbox)**, and **ES6+ JavaScript**.
 - **Backend (Proxy):** A **Python (HTTPServer)** proxy handles API authentication and bypasses CORS (Cross-Origin Resource Sharing) restrictions.
 - **Data Persistence:** Uses the browser's **localStorage** for privacy-centric asset tracking.
 - **Visualizations:** Custom **Canvas API** implementation for dynamic P/L distribution bars, donut charts for allocation, and line charts for history.
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3. Core Features & User Interface

The UI is designed with a "Live Cockpit" philosophy, featuring a high-contrast dark mode and real-time data synchronization.

3.1 Live Market Dashboard

The dashboard displays trending assets with 24-hour and 7-day price changes, as well as total market capitalization.

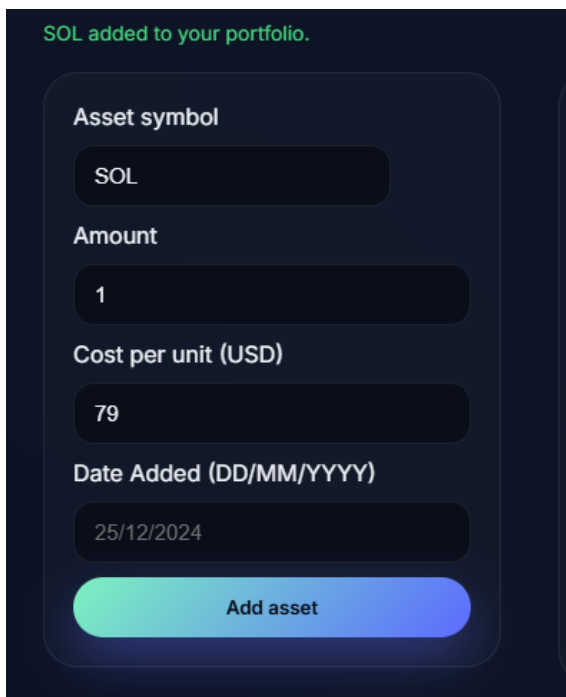


Markets		Portfolio timeline		
Asset	Price	24h	7d	Market cap
BTC Bitcoin	\$68,806.03	-2.74%	-9.85%	1.4T
ETH Ethereum	\$2,011.64	-5.47%	-12.59%	243B
SOL Solana	\$82.60	-5.75%	-18.04%	47B
BNB BNB	\$617.30	-3.56%	-19.77%	84B
XRP XRP	\$1.40	-3.71%	-13.60%	85B
DOGE Dogecoin	\$0.0928	-3.54%	-14.98%	16B
ADA Cardano	\$0.2621	-3.12%	-12.88%	9.5B
AVAX Avalanche	\$8.72	-3.95%	-14.56%	3.8B

Figure 1: Real-time market data dashboard. This view fetches live price, 24h/7d percentage changes, and market capitalization for major cryptocurrencies directly via the CoinMarketCap API proxy.

3.2 Portfolio Management & Input

Users can add assets by providing the symbol, amount, cost per unit, and the date of purchase. The application ensures data integrity through input validation.



SOL added to your portfolio.

Asset symbol

SOL

Amount

1

Cost per unit (USD)

79

Date Added (DD/MM/YYYY)

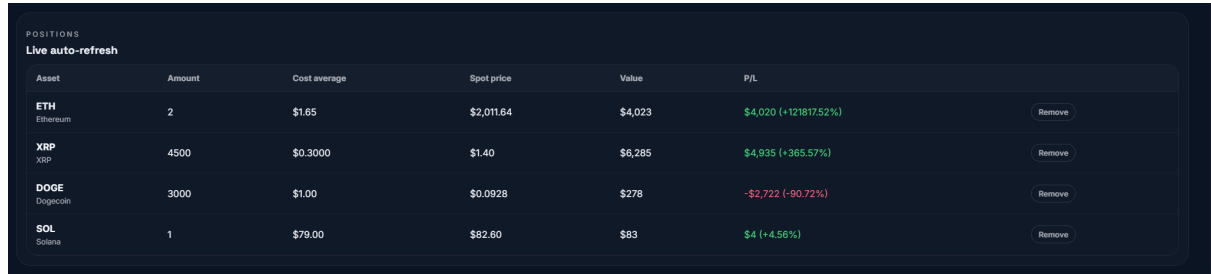
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Add asset

Figure 2: User entry interface for portfolio assets. The form allows users to input the symbol, quantity, unit cost, and historical date, ensuring accurate P/L calculations based on the entry price.

3.3 Live Asset Tracking

Once assets are added, the "Positions" table provides an auto-refreshing overview of each holding. It calculates the **Cost Average**, **Spot Price**, total **Value**, and the specific **P/L** for each individual asset.



Asset	Amount	Cost average	Spot price	Value	P/L	
ETH Ethereum	2	\$1.65	\$2,011.64	\$4,023	\$4,020 (+121817.52%)	<button>Remove</button>
XRP XRP	4500	\$0.3000	\$1.40	\$6,285	\$4,935 (+365.57%)	<button>Remove</button>
DOGE Dogecoin	3000	\$1.00	\$0.0928	\$278	-\$2,722 (-90.72%)	<button>Remove</button>
SOL Solana	1	\$79.00	\$82.60	\$83	\$4 (+4.56%)	<button>Remove</button>

Figure 3: *Dynamic holdings table. This component calculates real-time portfolio metrics including cost average, current market value, and total profit/loss (P/L) for each individual position.*

3.4 Advanced Analytics & Visualization

The analytics suite provides a macro view of the portfolio's health using dynamic charts:

- **P/L Distribution:** A horizontal bar chart visualizing the performance of each asset.
- **Allocation Chart:** A donut chart illustrating portfolio diversification.

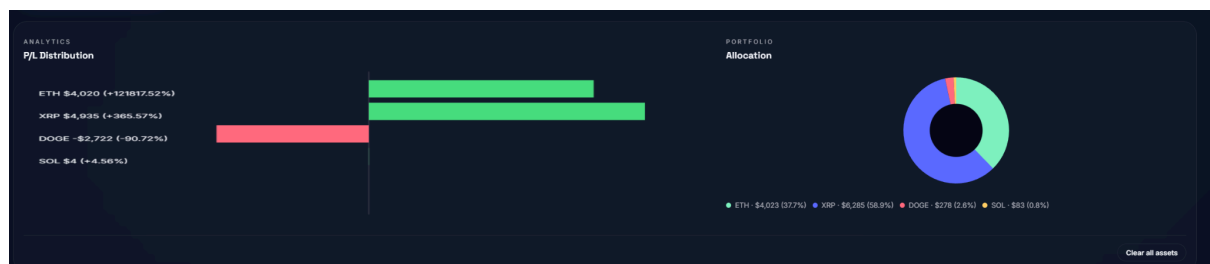


Figure 4: *Portfolio analytics overview. On the left, a custom Canvas API bar chart shows the distribution of profit/loss; on the right, a donut chart visualizes the asset allocation and diversification of the total portfolio.*

4. Backend Implementation (The Proxy Server)

To protect the `CMC_API_KEY`, the project includes a `server.py` component:

- **Security:** The API key is stored in `coinmarket.ini` and never exposed to the client-side code.
 - **Endpoint:** It maps local requests from `http://127.0.0.1:5050/api/quotes` to the official CoinMarketCap Pro API.
 - **Headers:** Automatically injects the required `X-CMC_PRO_API_KEY` for every request.
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5. Challenges & Solutions

Challenge	Solution
CORS Restrictions	Implemented a Python Proxy server to handle headers and requests securely.
Data Privacy	Used <code>localStorage</code> instead of a central database to keep user data on their own machine.
UI Responsiveness	Used <code>clamp()</code> and <code>grid</code> in CSS to ensure the dashboard works on mobile and ultra-wide screens.
High Precision Math	Implemented custom formatting functions for crypto prices (e.g., showing 4-8 decimals for low-value coins).

6. Future Roadmap

- Firestore Integration:** Optional cloud sync for multi-device support.
- WebSockets:** Real-time "ticker" style updates without page refreshing.
- Export Options:** PDF and CSV reporting for financial tax records.

7. Conclusion

AssetPulse successfully bridges the gap between complex financial data and user-friendly design. It demonstrates a full-stack proficiency in handling third-party APIs, asynchronous JavaScript, and real-time data visualization. This project showcases the ability to turn raw API data into actionable financial insights using a modern and secure architecture.